

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis

Internship Report — Gigalis (Pays de la Loire)

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Abstract

This report presents the work carried out during a data-science internship for **Gigalis**, a public-interest group and central purchasing body in the Pays de la Loire region. The goal is to move the analysis of public digital procurement from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). The central methodological obstacle is that the French procurement publication system (BOAMP) contains *no explicit field* marking whether a call for tenders renews an earlier contract. The renewal relationship must therefore be reconstructed from observable signals. We build an auditable, reproducible *proxy* renewal event using Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, and we use it as the event variable for survival analysis. On 1,100 eligible source contracts the linking method recovers a plausible renewal for 697 of them (**63.4%**), against an 11.3% lexical baseline. Survival modeling on this dataset gives a global Kaplan–Meier median survival of **48 months**, a Cox proportional-hazards concordance of **0.6537**, and a best-fitting log-normal parametric model. We report these results honestly together with their limitations: the renewal event is a proxy, buyer identification is weak, and two originally planned components (a trained NLP classifier and change-point detection) are not yet implemented.

1 Introduction and Problem Statement

1.1 Context

Gigalis is a *groupement d'intérêt public* that operates a central purchasing body for digital services in the Pays de la Loire region. As part of building a “data control tower”, the organisation wants to anticipate when its digital contracts (software, IT services, telecommunications, security) are likely to come up for renewal, rather than only describing past activity. The practical value is operational: a reliable renewal forecast helps plan re-tendering workloads and negotiate from a stronger position.

1.2 The core difficulty

Public procurement in France is published through two main open-data channels: **BOAMP** (*Bulletin officiel des annonces des marchés publics*, the tender and award notices) and **DECP** (*Données essentielles de la commande publique*, the awarded-contract records). Neither source contains a *renewal link*: a notice does not state “this tender replaces contract X that is expiring”. Yet the renewal event is exactly the quantity we want to model. The problem is therefore twofold:

1. **Event reconstruction.** Infer, from text, classification, timing, and buyer identity, which later notice is the renewal of an earlier contract — producing a binary **event** variable and a survival time.

2. **Survival modeling.** Given this reconstructed event, estimate how the probability of renewal evolves over time and which contract characteristics drive it.

1.3 Research question and scope

Given a digital procurement contract, when is it likely to be renewed or re-tendered, and which factors influence that timing? The study is deliberately scoped to keep the data tractable and the modeling honest:

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024.
- **Official modeling path: BOAMP-only.** DECP is explored as a richer but non-linkable source and kept for future enrichment (Section 2).

2 Data and Scope

2.1 Sources

BOAMP (official path). BOAMP is queried through the Opendatasoft Explore API (dataset `boamp`). After filtering to the target CPV divisions, territory, and period, the acquisition yields **3,181 notices**. One technical subtlety is that CPV codes are embedded inside a JSON-encoded field in *two* different serialisations: the legacy BOAMP format (notices up to 2023) and the EU eForms/UBL format (2024 onward); both are parsed and cross-checked. From the notices, the APPEL_OFFRE (call for tenders) records define the survival population: **1,933 source contracts**.

DECP (exploratory only). DECP is obtained from the consolidated tabular file on `data.gouv.fr` (~210 MB). It contains **3,039** digital contracts for the same scope and is far better populated than BOAMP (100% SIRET identifiers, 95.6% amounts, 99.4% durations). However, BOAMP and DECP *do not share a reliable common key*, so a robust contract-level merge is not achievable. DECP is therefore documented and retained as a future enrichment path, but it is **not** part of the official BOAMP-only modeling handoff.

2.2 Taxonomy

To analyse renewal by technology family, each contract is tagged with one of ten categories (CAT01–CAT10: Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.). This taxonomy is a **deterministic rule-based mapping** from CPV codes and description keywords — it is *not* a trained classifier. This is an important honesty point: the project plan originally foresaw a learned NLP classifier, but the implemented component is rule-based.

2.3 Data quality

The data-quality assessment drives every downstream modeling choice (Table 1). The decisive weaknesses are buyer identification and duration granularity.

Table 1: Key BOAMP data-quality findings and their modeling consequence.

Finding	Value (BOAMP)	Consequence
Buyer SIRET coverage	~9%	Grouping relies on normalised buyer <i>names</i> ; one entity may fragment into several keys.
Amount fill rate	~41–43%	Amount used cautiously, not as a core covariate.
Declared duration granularity	69% exactly 12/24/36/48 mo	Duration treated as a <i>covariate</i> , not as the survival target.

2.4 Dataset construction and missing-value handling

The project dataset is not a direct download of “contract renewals”. It is a constructed analytical dataset. BOAMP publishes individual notices, so the work first converts notice-level data into a contract-level table suitable for survival analysis.

1. **Extract and flatten.** The BOAMP API output is flattened into `boamp_full_flat.csv`: 3,181 notices and 30 raw/flattened fields.
2. **Clean without hiding uncertainty.** The cleaned file, `boamp_full_clean.csv`, keeps the original fields and adds controlled columns such as `buyer_key`, `cpv_clean`, `amount_clean`, `duration_clean`, and category labels.
3. **Define possible contracts.** Only the 1,933 APPEL_OFFRE notices are treated as source contracts, because these are the records that can later be renewed.
4. **Define the observable modeling cohort.** The final denominator keeps 1,100 contracts whose expected renewal window is observable before 2024-12-31. The remaining 833 are too recent to evaluate fairly.
5. **Build the survival handoff.** The final `boamp_phase2_survival.csv` file contains one row per eligible contract, with the event indicator, observed or censored duration, covariates, and link-confidence diagnostics.

The missing-value policy is deliberately conservative: raw fields are preserved, quality problems are flagged, and imputation is used only when the next algorithmic step requires a value. In particular:

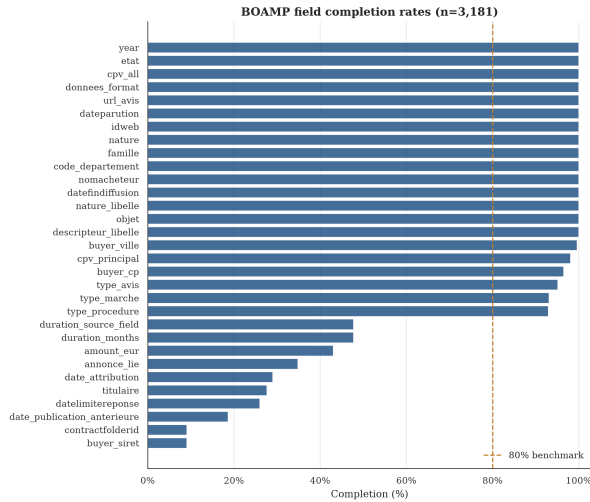
- **Buyer identity.** If a valid 14-digit SIRET exists, the buyer key is SIRET-based. Otherwise the pipeline uses a normalised buyer name. This is necessary for grouping, but it is also the main source of false negatives because spelling variants can split the same buyer.
- **Amounts.** BOAMP amounts are sparse: only 21.4% of call-for-tender notices have a valid `amount_clean`. Zero, tiny ($< 1,000$ EUR), and very large (at least 10M EUR) values are flagged and removed from the cleaned amount, while the raw amount remains stored. Amount is therefore not used as a core matching signal.
- **Durations.** Suspect durations outside 1–120 months are flagged. Missing or suspect durations are imputed to 48 months only for building the renewal search window; the binary `dur_was_imputed` flag records this. The declared duration is a covariate and search prior, not the survival target.
- **CPV codes.** CPV is cleaned into the full hierarchy (`cpv_div2`, `cpv_group3`, `cpv_class4`, `cpv_category5`) and used for taxonomy and matching. If CPV is missing in a candidate pair, it is excluded from the composite score and the remaining component weights are renormalized, rather than being given an arbitrary neutral value or causing an automatic rejection.

- **Renewal labels.** BOAMP has no explicit renewal flag. The **event** variable is reconstructed by linking later notices from the same buyer with compatible text, CPV, and timing. Thus **event=0** means no renewal was observed before the study end under the algorithm; it is not proof that the contract was never renewed.

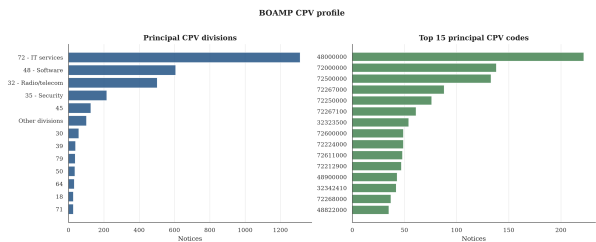
This distinction is crucial for explaining the model: missing amounts are a data-quality limitation, imputed durations are a controlled assumption, and right-censored contracts are valid survival observations rather than bad rows.

Table 2: Quick practical reading of common dataset cases.

Case	Example in the data	How to read it quickly
Missing amount	A BOAMP call for tender can have <code>amount_clean = NaN</code> .	The contract is still usable for linking. It only means amount is not reliable enough to use as a main signal.
Missing duration	<code>dur_was_imputed = True</code> , for example when a duration is set to 48 months.	The row stays in the dataset, but I know the renewal search window is based on an assumption.
Linked renewal	<code>event = 1</code> , with a filled <code>renewal_contract_id</code> .	The algorithm found a later notice that looks like a renewal; the observed duration is the gap to that later notice.
No observed renewal	<code>event = 0</code> , with <code>censoring_duration_months</code> filled.	This is not a failed row. It means the contract was observed until study end without an accepted renewal link.



(a) BOAMP field completeness.



(b) Distribution by CPV division.

Figure 1: BOAMP corpus profile: field completeness (left) shows the sparse amount/SIRET fields that constrain modeling; CPV distribution (right) is dominated by divisions 72 and 48.

3 Method

The pipeline has four stages: cleaning, renewal-event reconstruction, survival dataset construction, and survival modeling.

3.1 Cleaning

Raw notices are normalised: amounts are filtered for zero, suspiciously small, and ceiling-flagged values; durations are parsed and, when missing, imputed to a default (a binary `dur_was_imputed` flag records every imputation); buyer names are normalised into a `buyer_key`; and the taxonomy tags are applied. The cleaned corpus is the input to linking.

3.2 Renewal-event reconstruction (core contribution)

A renewal is reconstructed by matching each source `APPEL_OFFRE` to a later candidate notice from the same buyer.

Eligibility. A source contract is *eligible* only if its expected renewal window (start date plus declared duration) falls inside the 2015–2024 study period. Contracts whose renewal would necessarily occur after 2024-12-31 are right-censored *before* analysis. This reduces the 1,933 source contracts to **1,100 eligible** (833 censored upfront). Restricting evaluation to eligible contracts is what makes the linking rate meaningful rather than mechanically deflated.

Composite score. For an eligible source i and candidate j with a CPV on both sides, a composite similarity score combines four signals:

$$S(i, j) = 0.40 s_{\text{text}} + 0.25 s_{\text{cpv}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}. \quad (1)$$

When CPV is missing on either side it is excluded and the remaining weights are renormalized: $S = (0.40 s_{\text{text}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}})/0.75$.

- s_{text} : cosine similarity of multilingual Sentence-Transformer embeddings (`paraphrase-multilingual-MiniLM-L12-v2`) of the contract descriptions. This is the strongest discriminator and the main improvement over the lexical baseline.
- s_{cpv} : hierarchy-graded CPV compatibility following the official CPV nomenclature (1.0 same full 8-digit code, 0.8 category, 0.6 class, 0.4 group, 0.2 division, 0.0 different division; generic catch-all codes capped at 0.2; missing CPV excluded, weights renormalized) — a *scored* signal, not a hard filter, because buyers sometimes reclassify.
- s_{temp} : temporal proximity to the expected renewal date.
- s_{buyer} : buyer-identity agreement.

Hard filters (same buyer, candidate strictly later, minimum text similarity 0.20) prune the candidate set; the best-scoring candidate is then selected. Accepted links are graded into confidence tiers (low < 0.50 , medium 0.50–0.70, high ≥ 0.70), which supports the sensitivity analysis.

Score margin and a strict high-confidence flag. The absolute composite score says how good the chosen candidate is, but not how *separable* it is from its rivals. For each source we therefore also record the gap to the runner-up,

$$\Delta_i = S(i, j^{(1)}) - S(i, j^{(2)}), \quad (2)$$

where $j^{(1)}$ and $j^{(2)}$ are the best- and second-best-scoring candidates. A small Δ_i flags an ambiguous winner. Of the 697 links, 135 have a single candidate (margin undefined) and 562 have ≥ 2 ; among the latter the median separation is comfortable but 273 links sit below $\Delta = 0.05$ (Figure 2). We

define a conservative `high_confidence_strict` indicator that requires *both* a high absolute score *and* a clear margin,

$$\text{event} = 1 \wedge S \geq 0.70 \wedge \Delta \geq 0.05,$$

which 90 links satisfy. Single-candidate links have an undefined margin and are deliberately *not* auto-promoted to strict; they are tracked separately so “clear winner among several candidates” is never conflated with “only candidate available”. This flag is *not* a new ground truth — it is a conservative sensitivity definition used to bound, not correct, the renewal signal (Model C in Section 4).

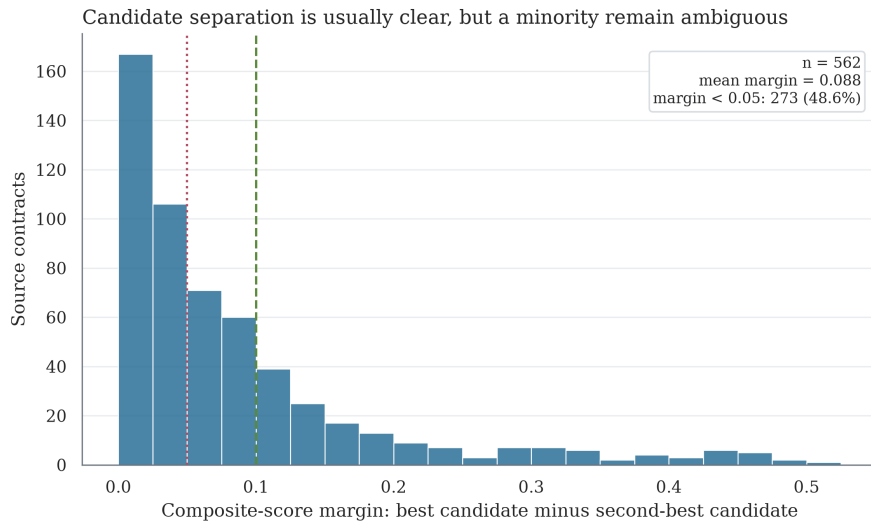


Figure 2: Best-minus-second-best composite-score margin for multi-candidate links. The dotted line marks the strict threshold $\Delta = 0.05$ and the dashed line $\Delta = 0.10$; the mass below 0.05 is the ambiguous tail removed under Model C.

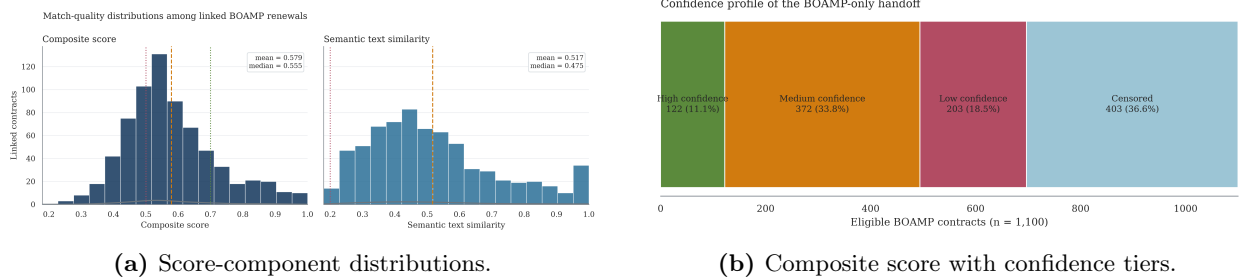


Figure 3: Linking diagnostics. The composite score (right) spreads accepted links across confidence tiers, motivating the conservative sensitivity variant in Section 4.

3.3 Survival dataset construction

Each eligible contract becomes one survival row:

- `event` = 1 if a plausible renewal link was found; the survival time is the observed gap (months) to the renewal.
- `event` = 0 if no link was found before study end; the survival time is the censoring time from start date to 2024-12-31.

The result, `boamp_phase2_survival.csv`, has one row per eligible contract with the following covariates:

- `declared_duration_months`
- `dur_was_imputed`
- `start_year`
- category and procedure type

3.4 Survival models

Using this dataset we fit: (i) non-parametric **Kaplan–Meier** estimators, globally and stratified by category, declared-duration group, and imputation flag, with **log-rank** tests for differences; (ii) a **Cox proportional-hazards** model (univariate and multivariate); and (iii) **parametric** accelerated-failure-time models (Weibull, log-normal, log-logistic) compared by AIC. A sensitivity analysis re-runs the whole modeling chain on a stricter, high-confidence subset of links.

4 Results

4.1 Renewal linking

The method links **697 of 1,100** eligible source contracts, a linking rate of **63.4%**, versus **11.3%** for the lexical (Jaccard/TF-IDF) baseline (Table 3). The gain comes from two changes: semantic embeddings instead of lexical overlap, and treating CPV as a graded score rather than a hard grouping constraint.

Table 3: Renewal-linking outcome on BOAMP source contracts.

Quantity	Value
Source APPEL_OFFRE contracts	1,933
Right-censored before analysis	833
Eligible source contracts	1,100
Linked renewals (event = 1)	697 (63.4%)
Right-censored (event = 0)	403 (36.6%)
Lexical baseline linking rate	11.3%

Renewal rates vary by technology family (Table 4): Software & Applications (70.8%) and IT Services & Consulting (67.6%) renew most often, while Data & AI (41.2%) and IT Hardware (47.6%) renew least. Small categories (e.g. GIS & Mapping, $n = 5$) are statistically underpowered and should be read with caution.

Table 4: Event rate by category on the 1,100 eligible contracts.

Category	n	events	rate %
IT Services & Consulting	312	211	67.6
Software & Applications	209	148	70.8
Telecom & Networks	204	120	58.8
Cybersecurity	121	76	62.8
Unknown	108	58	53.7
Digital Workplace & Collaboration	45	29	64.4
Data & AI	34	14	41.2
IT Maintenance & Support	22	14	63.6
IT Hardware & Equipment	21	10	47.6
Cloud & Infrastructure	19	13	68.4
GIS & Mapping	5	4	80.0

4.2 Survival analysis

The global Kaplan–Meier estimator gives a **median survival of 48 months** — consistent with the prevalence of 48-month framework contracts. Stratified curves (Figure 4) separate clearly by declared-duration group and by imputation flag, both highly significant under the log-rank test ($p < 0.001$, Table 5); the difference *across categories* is not significant ($p = 0.15$), indicating that timing is driven more by contract duration than by technology family.

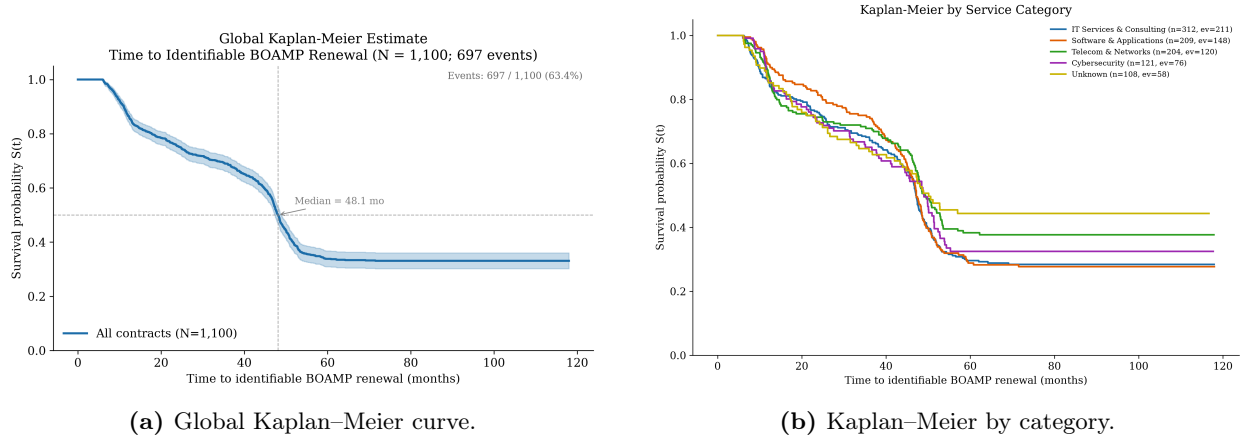


Figure 4: Survival to renewal. The global median is 48 months; category curves overlap substantially (log-rank $p = 0.15$).

Table 5: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	6.75	0.150	4
Duration was imputed	24.41	< 0.001	1
Declared-duration group	62.43	< 0.001	2
Procedure type (collapsed)	2.10	0.349	2

Cox model. The multivariate Cox model reaches an in-sample concordance of **0.6537**. Two predictors are statistically significant (Table 6): a longer declared duration lowers the renewal hazard ($HR = 0.990$ per month, $p < 0.001$), and later start years carry a higher hazard ($HR = 1.097$ per year, $p < 0.001$). Neither technology category nor procedure type is significant once duration and start year are controlled for, echoing the log-rank result.

Table 6: Cox proportional-hazards model: significant covariates (multivariate).

Covariate	HR	95% CI	p
declared_duration_months	0.990	[0.986, 0.995]	< 0.001
start_year	1.097	[1.053, 1.142]	< 0.001
dur_was_imputed	0.864	[0.718, 1.040]	0.123
Category / procedure terms	not significant ($p > 0.1$)		

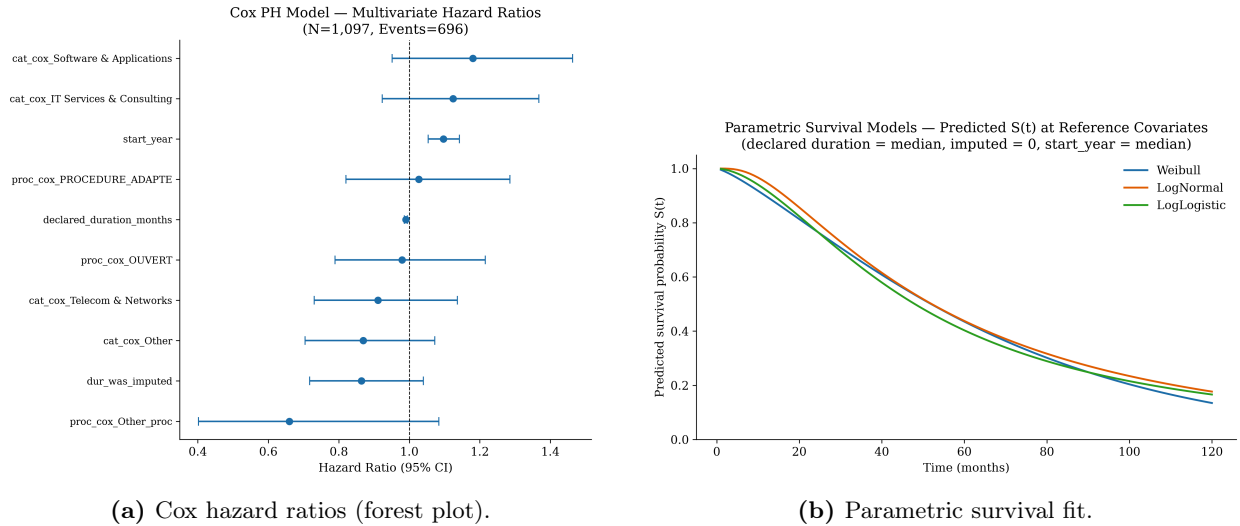


Figure 5: Cox effect sizes (left) and parametric fit (right). Confidence intervals crossing 1.0 confirm the non-significant category/procedure terms.

Parametric models. Among accelerated-failure-time models, the **log-normal** fits best (AIC = 7,119.8, concordance 0.6723), ahead of log-logistic (7,194.3) and Weibull (7,320.3). The right-skewed renewal-gap distribution is therefore better captured by a log-normal than by a Weibull hazard.

Table 7: Parametric model comparison (lower AIC is better).

Model	AIC	Concordance
Log-normal	7,119.8	0.6723
Log-logistic	7,194.3	0.6771
Weibull	7,320.3	0.6355

Sensitivity. We stress-test the event definition with two stricter variants (Table 8). Model B keeps only composite ≥ 0.50 links, dropping events from 697 to 494 (44.9%) so that the KM median is no longer reached, with Cox concordance moving modestly from 0.6537 to 0.6041. Model C is far more aggressive: it keeps `event` = 1 only for the 90 `high_confidence_strict` links (high score *and* a clear margin over the runner-up, Eq. 2) and *censors all other originally linked contracts at study end*. The event rate collapses to 8.2% and the KM median is no longer reached, so the apparent *level* of renewal risk falls sharply — as expected when most of the proxy events are removed. Crucially, the Cox concordance barely moves (0.6179), meaning the *ranking* of renewal risk by covariate is robust even under the strictest definition. Model C is a conservative lower bound, not a corrected ground-truth estimate; the qualitative conclusions are stable across all three variants, which is the purpose of the check.

Table 8: Sensitivity to the event definition. Model C requires both a high composite score (≥ 0.70) and a clear margin over the second-best candidate ($\Delta \geq 0.05$); “—” marks a KM median that is never reached.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	697	63.4	48	0.6537
B — conservative (≥ 0.50)	494	44.9	—	0.6041
C — strict high-confidence	90	8.2	—	0.6179

5 Limitations and Honest Appraisal

The results are useful but bounded, and the boundaries matter:

- **The event is a proxy.** `event = 1` means “a plausible renewal link was found”, not a legally certified renewal. The pipeline is designed to be transparent and auditable, not to prove renewal.
- **Weak buyer identification.** Only $\sim 13\%$ of unique buyers carry a SIRET; the rest are matched by normalised name, which can split one public entity (e.g. “Ville de Nantes” vs “Nantes Métropole”) into several keys and create false negatives. The SIRET field appears *only* in 2024 eForms notices, which are all censored upfront, so the modeled cohort is matched by name alone — SIRET enrichment can help future data but not the historical window.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously despite the semantic embeddings.
- **Structural censoring.** Contracts started in 2021–2024 are largely censored, since a typical 48-month contract would renew after the 2024-12-31 cut-off; the positive `start_year` hazard partly reflects this.
- **Modest discrimination.** A concordance around 0.65 means useful but far-from-perfect ranking; predictions should inform planning, not replace judgement.

Not implemented. For full transparency, two components of the original project plan are *not* part of this study: a **trained NLP classifier** (the taxonomy here is rule-based) and **change-point / trend detection**. The DECP linking remains at the lexical baseline only; the Sentence-Transformer method was applied to BOAMP, not yet to DECP. Exploratory DECP renewal links and a combined `unified_survival.csv` (3,512 BOAMP+DECP rows) do exist in the repository, but they are unvalidated and are not part of the official BOAMP-only results reported here.

6 Conclusion and Future Work

This internship delivered an end-to-end, reproducible pipeline that turns raw BOAMP notices into a survival-ready dataset and a first predictive view of contract renewal for Gigalis. The key methodological contribution is the reconstruction of a renewal event from unlabelled procurement notices using Sentence-Transformer embeddings, lifting the linking rate from 11.3% to 63.4% on eligible contracts. The survival analysis shows that renewal timing is driven mainly by declared contract duration (median survival 48 months) rather than by technology family, with a log-normal model giving the best parametric fit.

Future work follows directly from the limitations: (i) apply the Sentence-Transformer linking method to DECP and attempt a BOAMP–DECP merge for amount/SIRET enrichment; (ii) strengthen buyer resolution using SIRET and a controlled directory of regional public entities; (iii) replace the rule-based taxonomy with a trained text classifier; and (iv) add change-point and trend detection to monitor shifts in procurement behaviour over time.